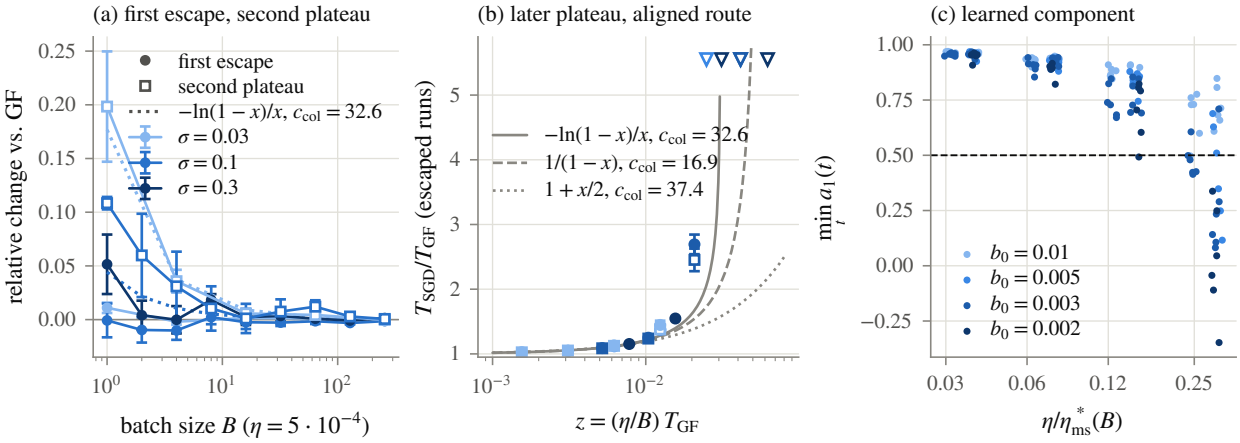
(a) first escape, second plateau
relative change vs. GF
batch size B ($\eta = 5 \cdot 10^{-4}$)
first escape
second plateau
$-\ln(1-x)/x$, $c_{\mathrm{col}} = 32.6$
$\sigma = 0.03$
$\sigma = 0.1$
$\sigma = 0.3$
(b) later plateau, aligned route
$T_{\mathrm{SGD}}/T_{\mathrm{GF}}$ (escaped runs)
$z = (\eta/B)\, T_{\mathrm{GF}}$
$-\ln(1-x)/x$, $c_{\mathrm{col}} = 32.6$
$1/(1-x)$, $c_{\mathrm{col}} = 16.9$
$1 + x/2$, $c_{\mathrm{col}} = 37.4$
(c) learned component
$\min_t a_1(t)$
$\eta/\eta^*_{\mathrm{ms}}(B)$
$b_0 = 0.01$
$b_0 = 0.005$
$b_0 = 0.003$
$b_0 = 0.002$